

Submission & Reviewer Matching

Weekly update from Qiang

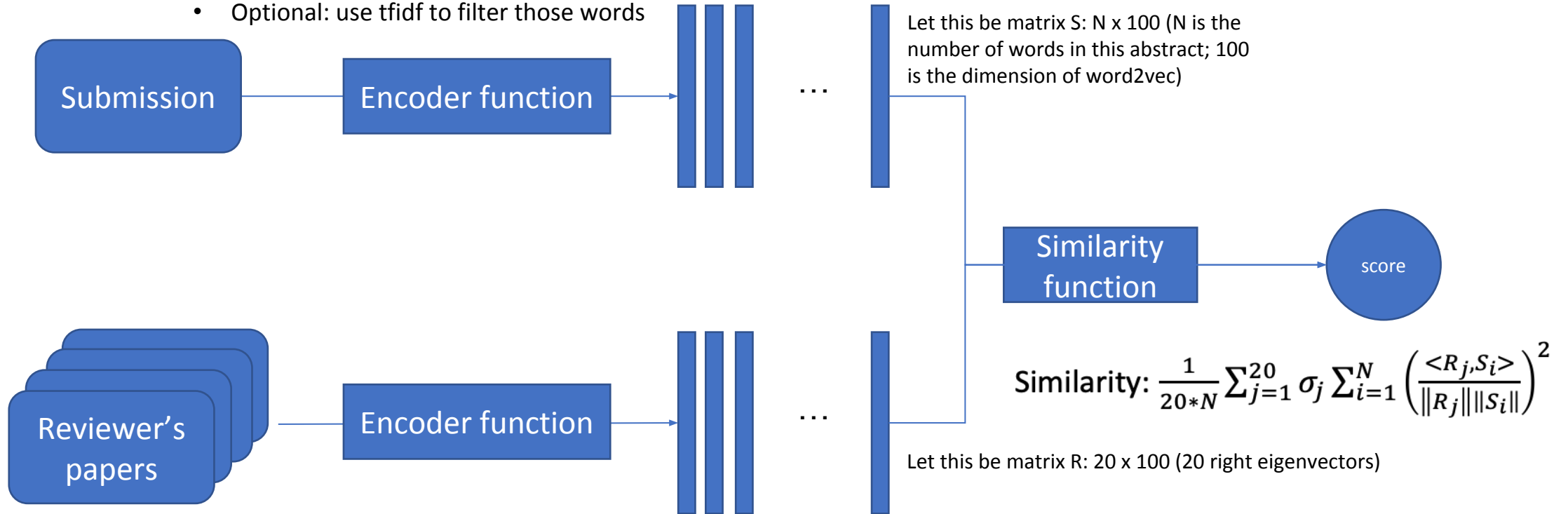
07/23/2019

Update

1. Initial try of TFIDF
2. Issues of our evaluation metric
3. Word embeddings
4. Topic weights

What we have now

- Pre-train word2vec on all the papers in our corpus
- Stack all the word vectors from the abstract of the submission
- Optional: use tfidf to filter those words

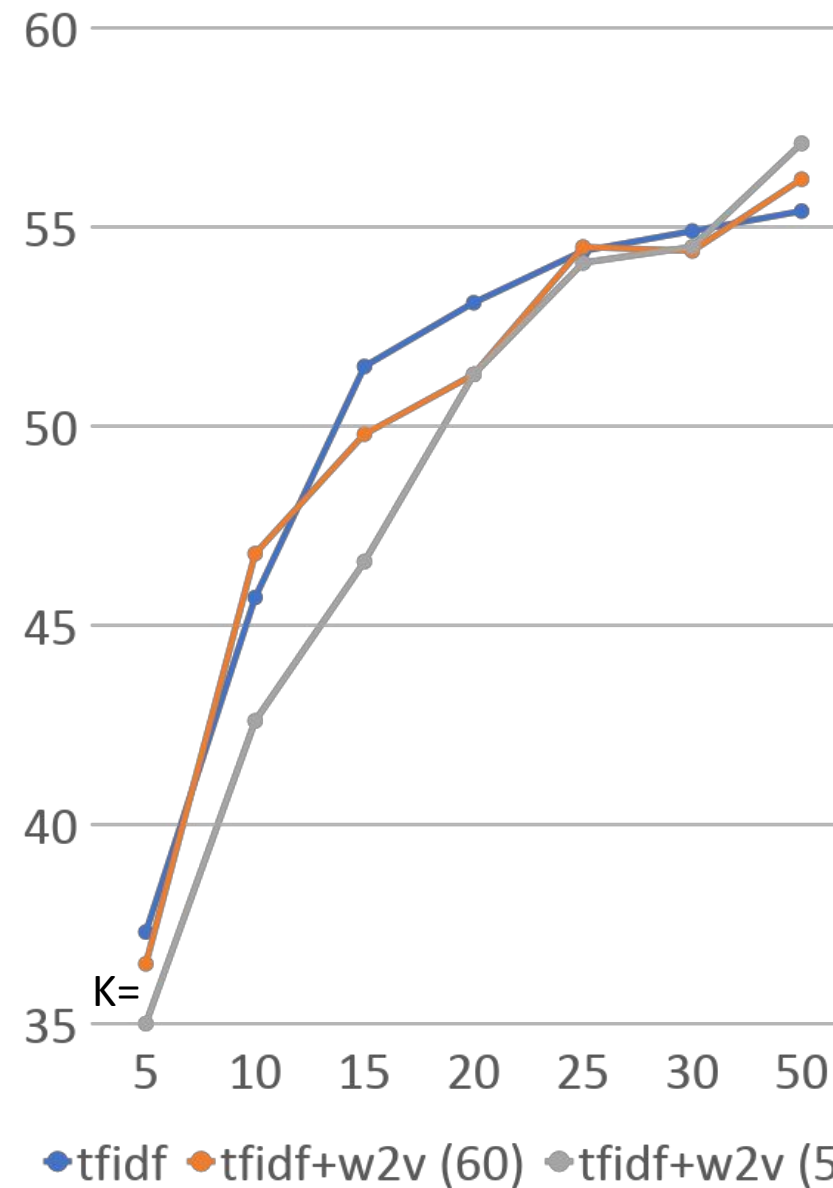
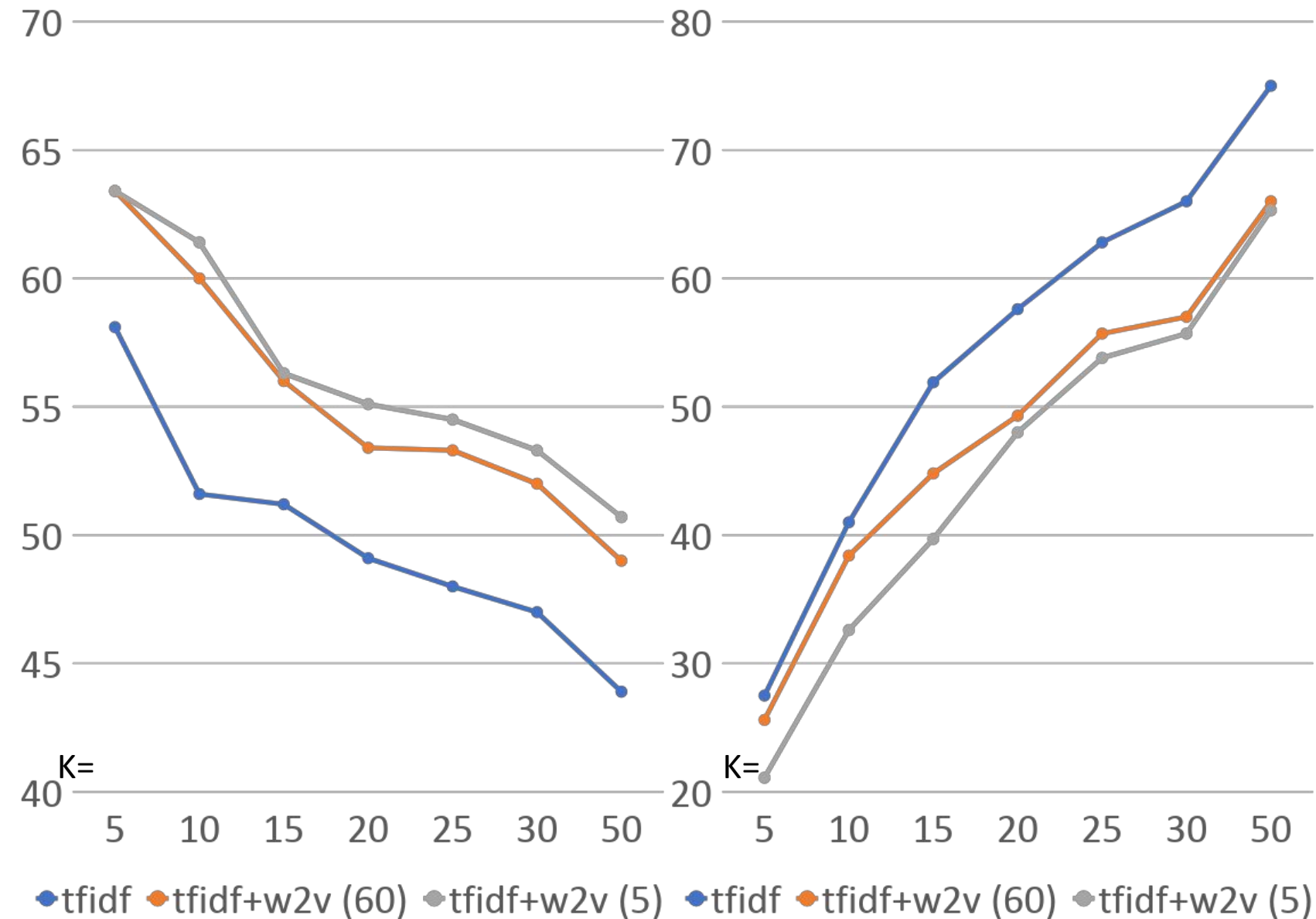


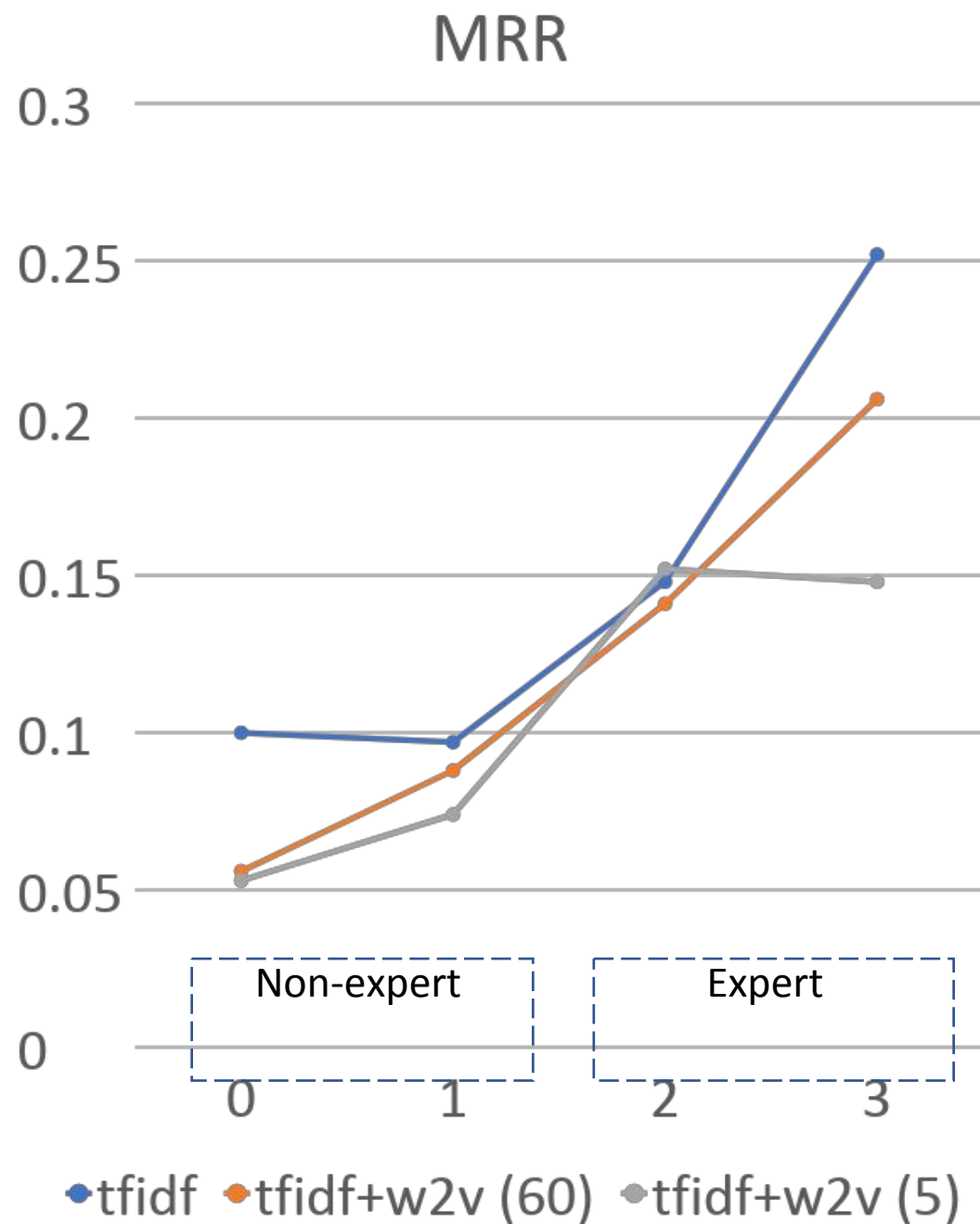
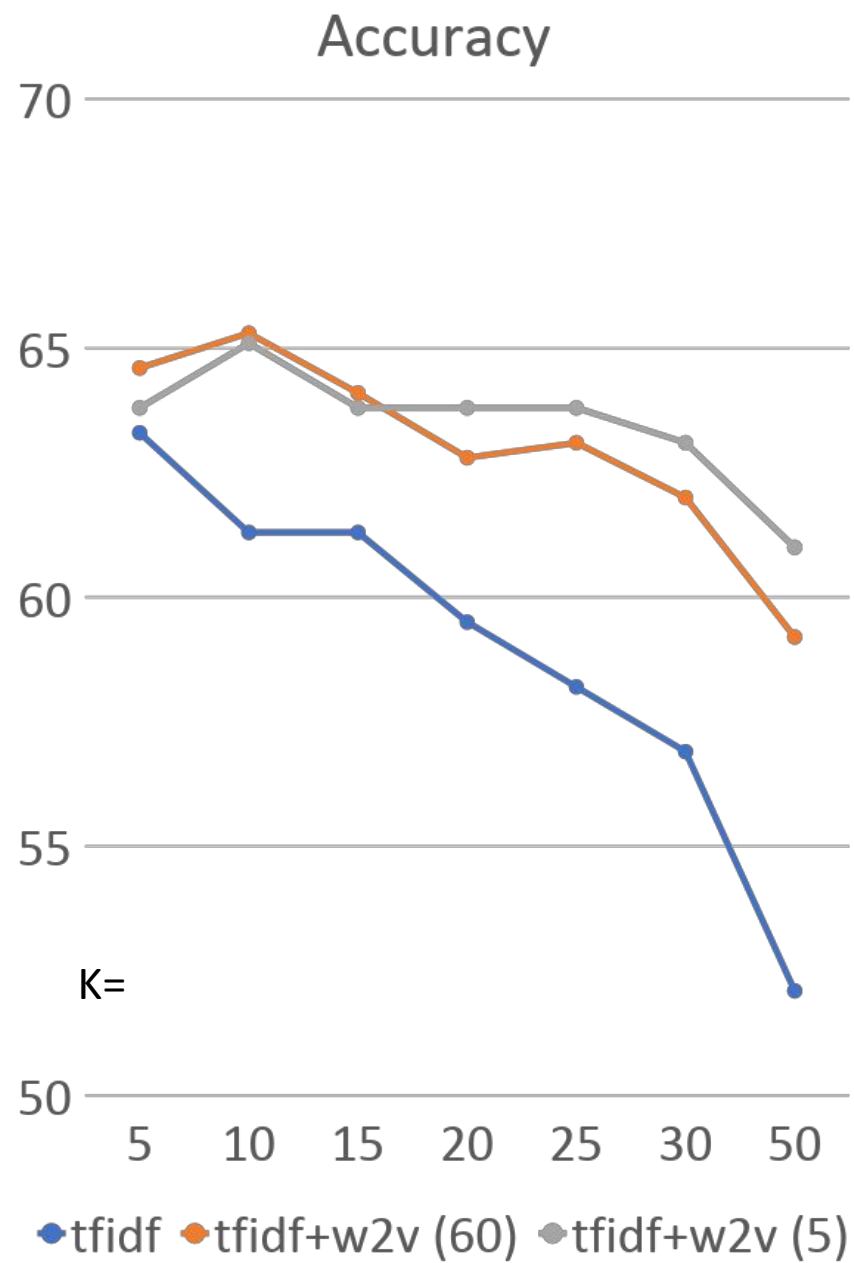
- Pre-define a weight vector for papers in specific years (putting more weight on more recent papers)
- Stack all the word vectors from the abstract of the papers of one particular review
- Perform SVD and obtain the first 20 right eigenvectors
- Optional: use tfidf to filter those words

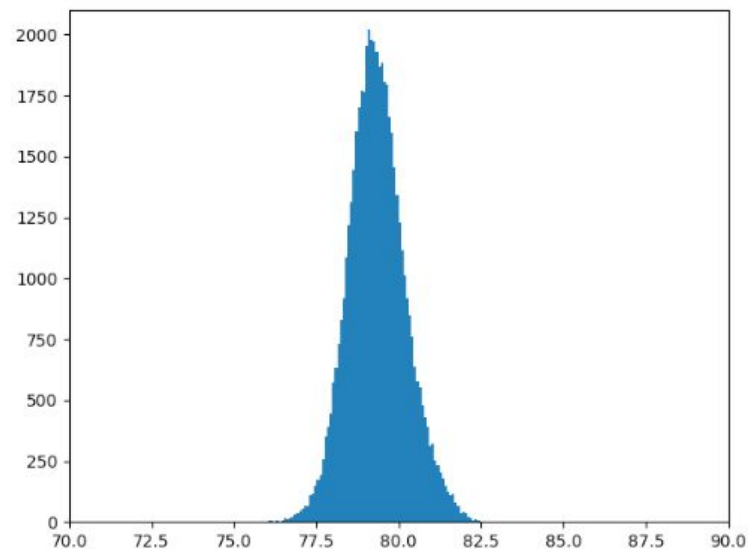
Precision

Recall

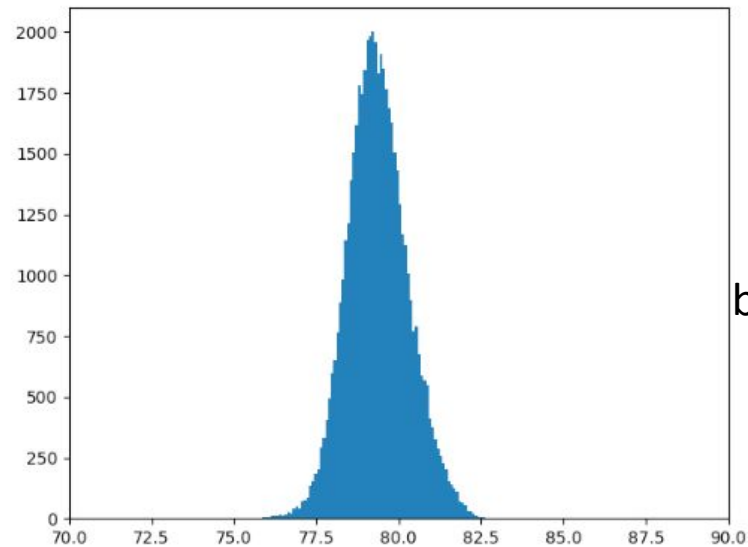
F1





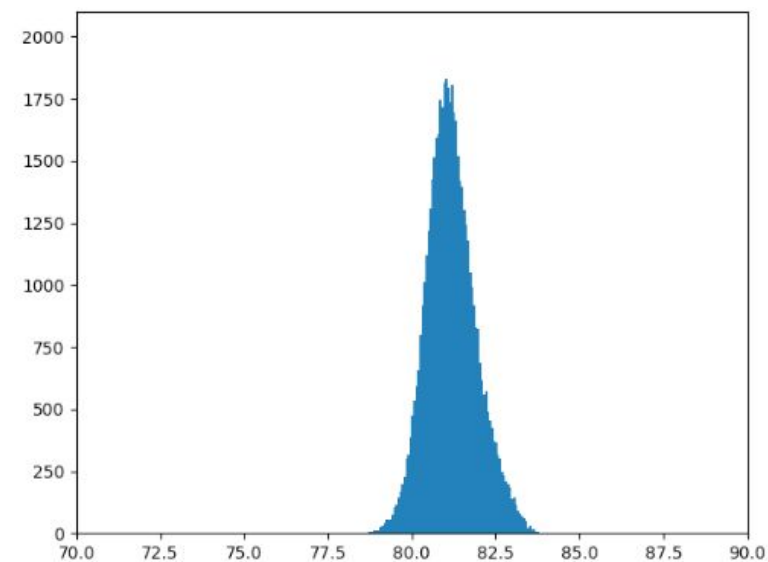


baseline



baseline +TFIDF

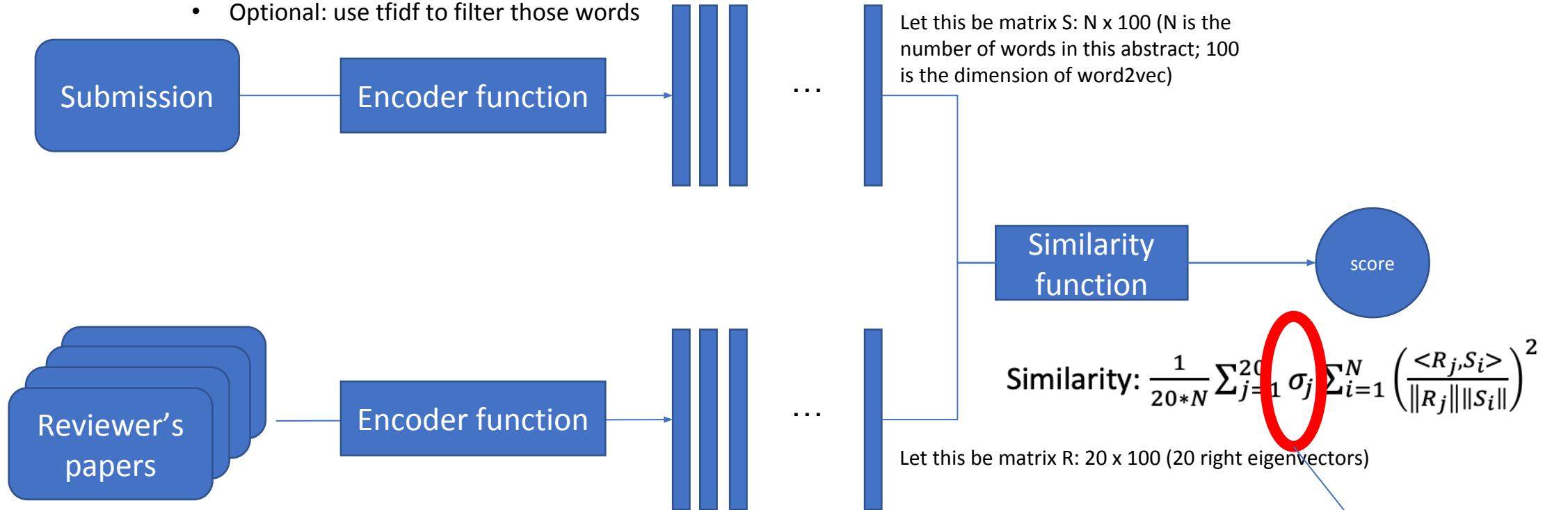
baseline
+TFIDF
+word2vec
Topic#=60



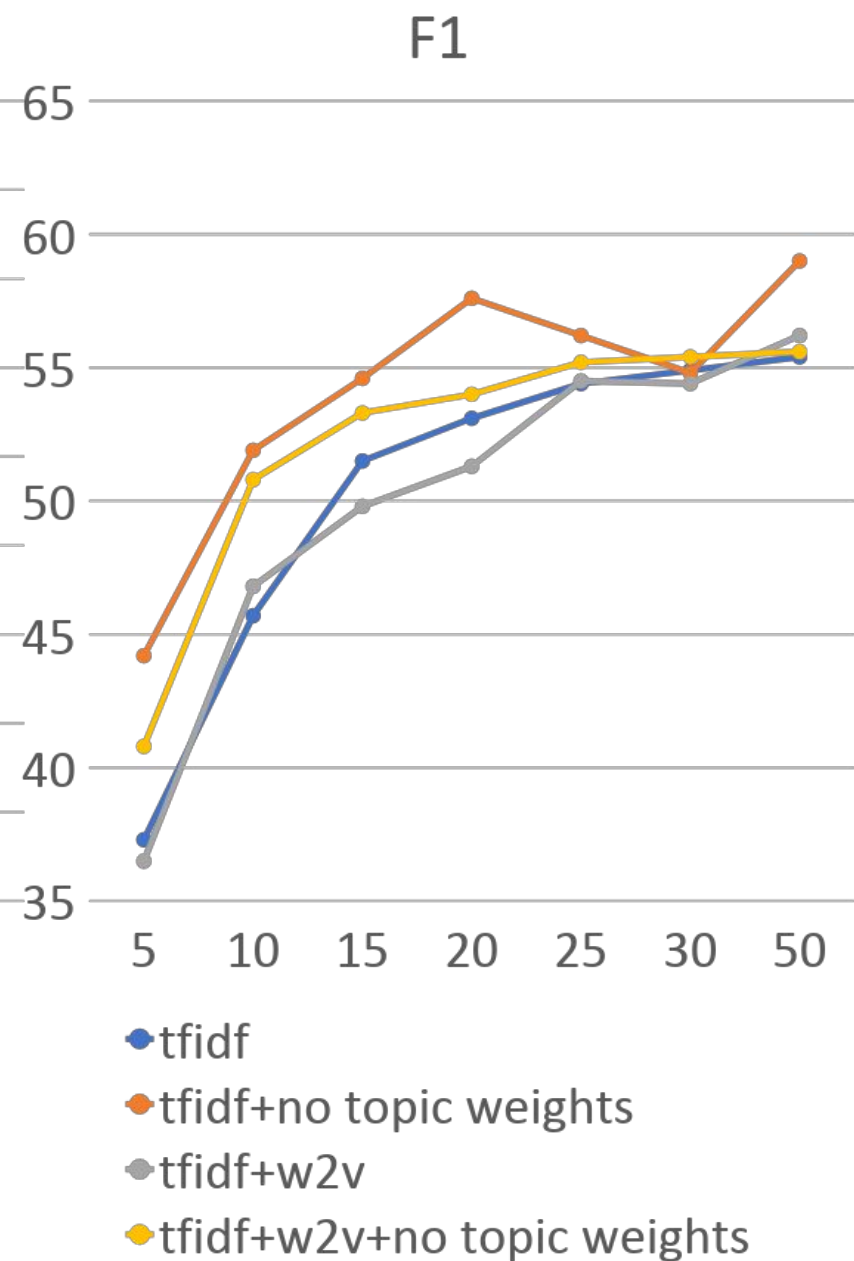
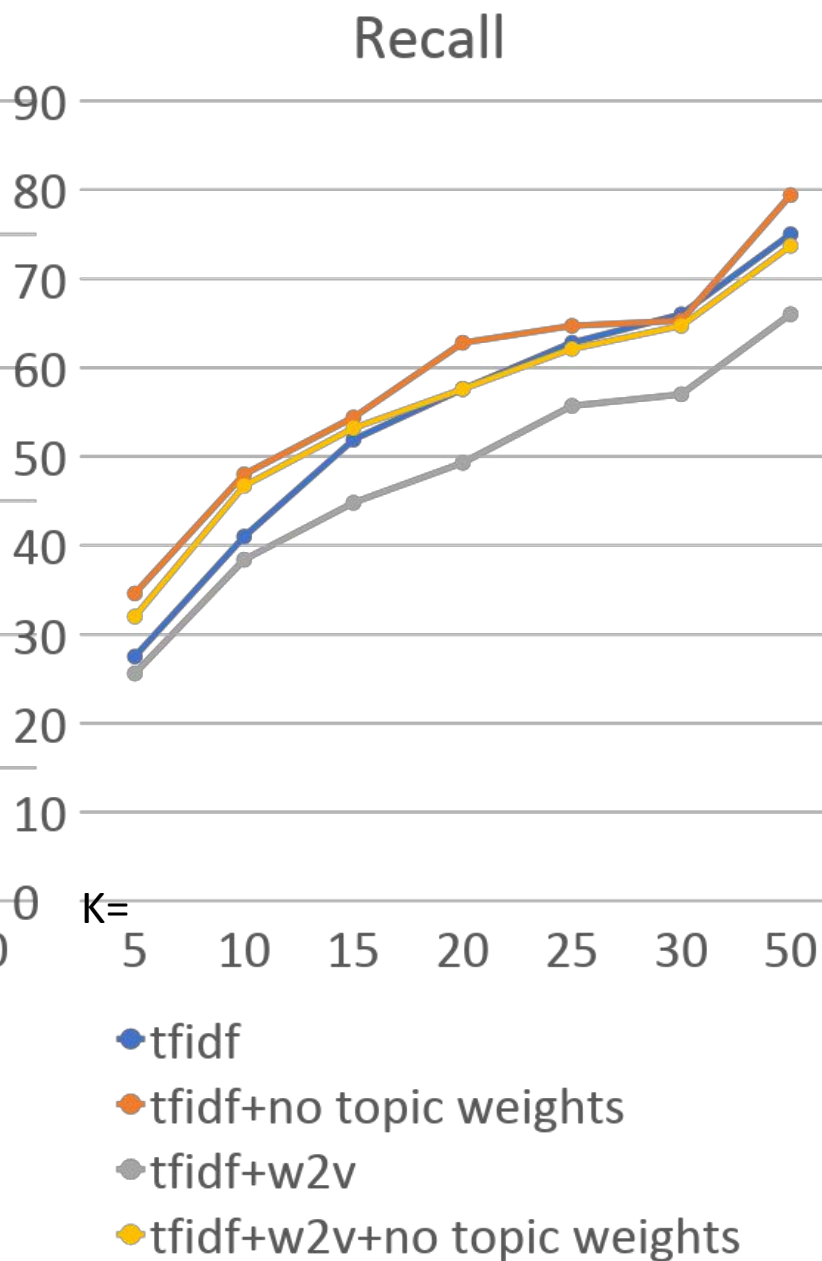
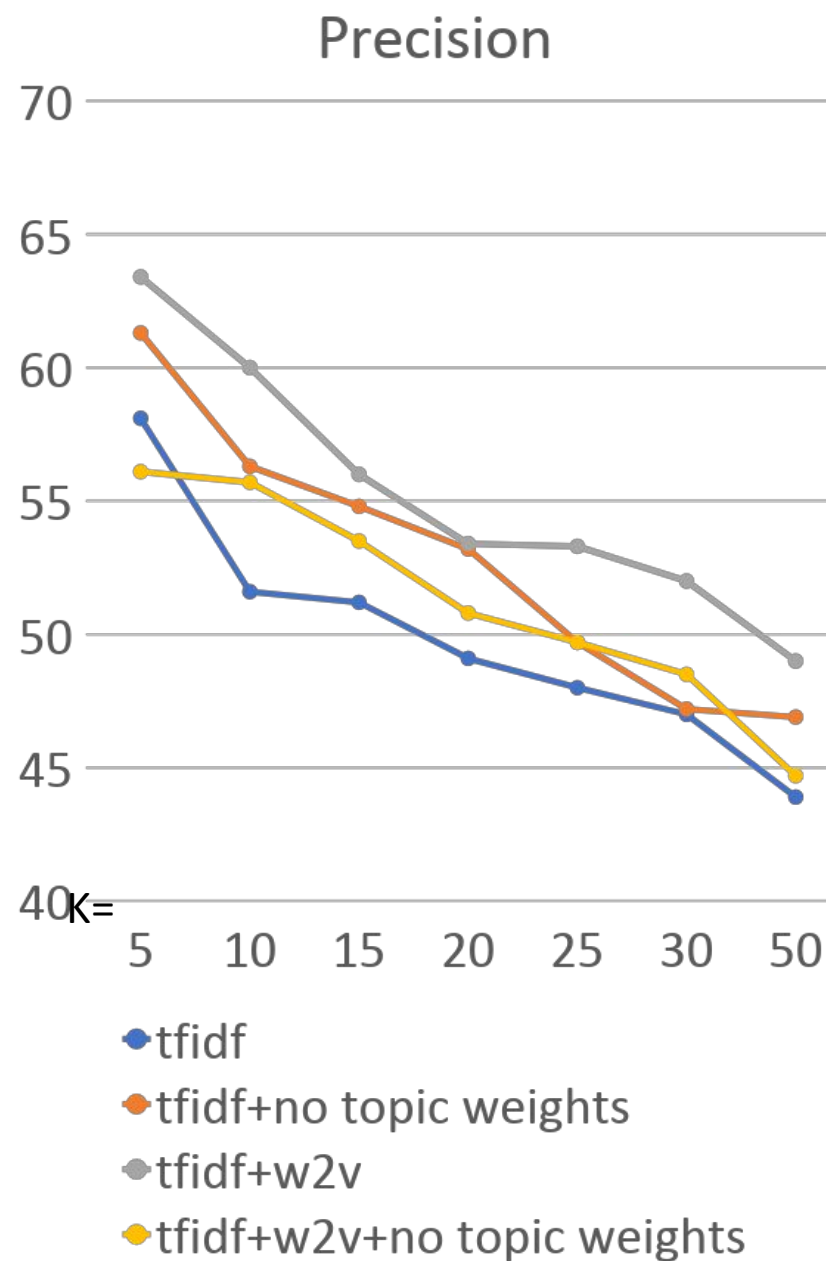
The concentration issue wasn't solved by TFIDF or word2vec.

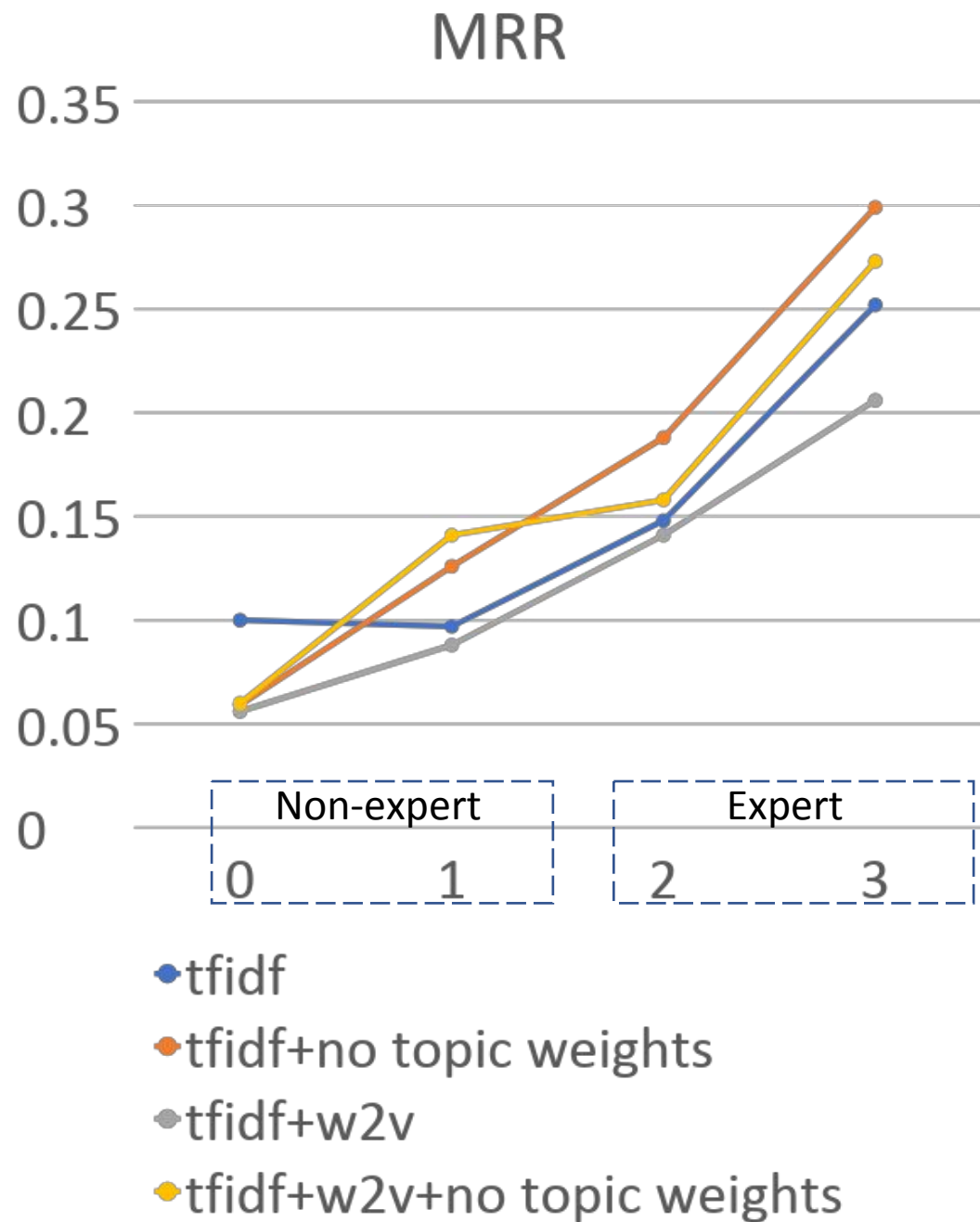
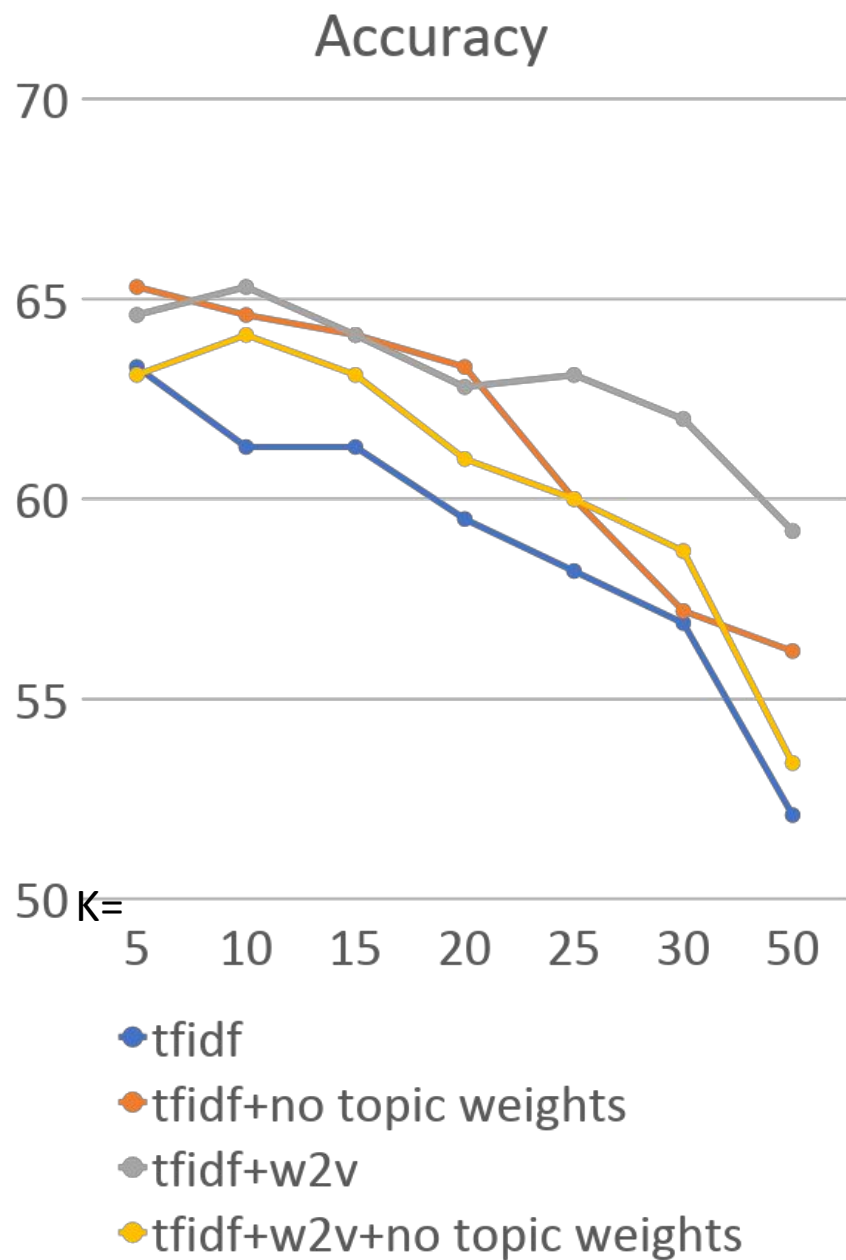
What we have now

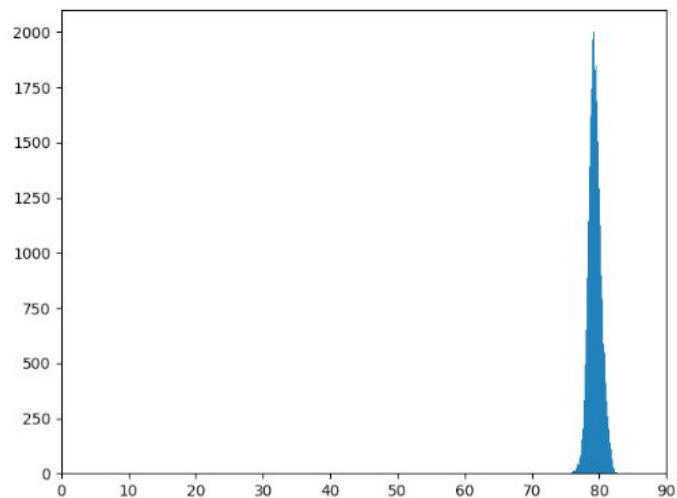
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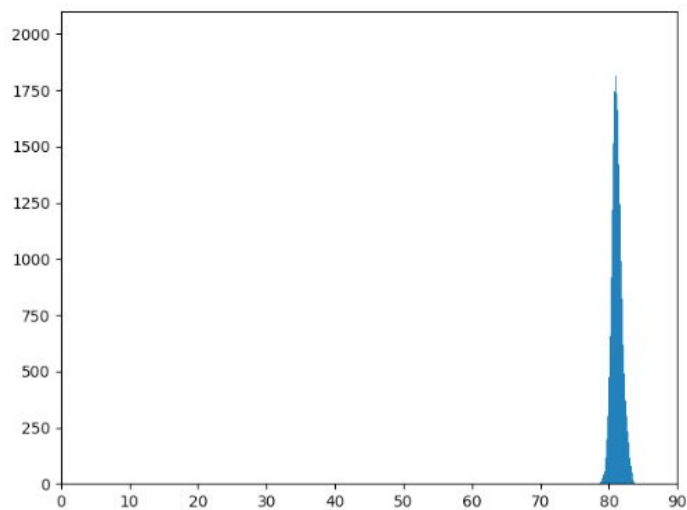
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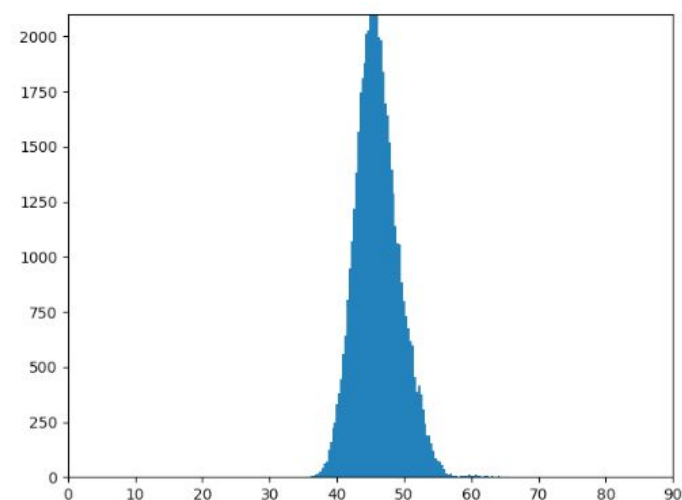


Old distribution
of scores

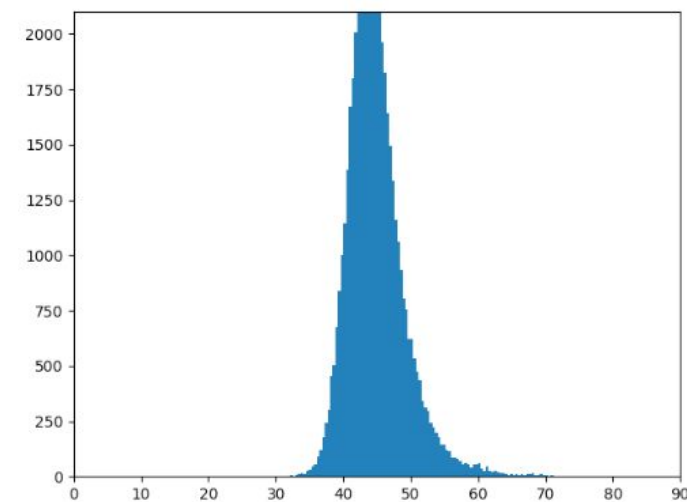


In domain w2v

Off-the-shelf w2v



New distribution of
scores (i.e., don't use
topic weights)



Summary

- Our in-domain word2vec isn't better than off-the-shelf word2vec. It means
 - Either we didn't train it properly
 - Or we shouldn't train it on papers
- Best system now: tfidf+no topic weights
 - Precision+3.2%, recall+7.1%, F1+6.9%, Acc+2%, MRR+4.7%
- “No topic weights” leads to a wider distribution.
- But it's unclear why no topic weights didn't help with off-the-shelf word2vec.
- ELMo failed to produce reasonable results in our current implementation.